Polynomial Families of Fourth-Power Identities with Variable Coefficients

Derivations, Generalizations, and Related Constructions

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Abstract

We study polynomial identities of the form

$$N(p,q)a(p,q)^4 + b(p,q)^4 + c(p,q)^4 = d(p,q)^4,$$

where the coefficient N and the four bases may depend on integer parameters. Two explicit identities are placed in coefficient-cancellation templates built from

$$(X+Y)^4 - (X-Y)^4 = 8XY(X^2+Y^2).$$

We prove both templates, recover the two starting identities as special cases, and give further examples. A binomial-expansion family is also included. The distinction between a polynomial identity, a solution for a fixed coefficient, and a novelty claim is made explicit.

Keywords: quartic Diophantine identities; polynomial identities; coefficient comparison; fourth powers.
MSC 2020: 11D25, 11D41.

1 Scope and starting identities

The purpose of this note is to give explicit algebraic templates for fourth-power identities, prove the templates, and record further specializations. The formulas do not solve the general problem of representing a fixed integer as a signed sum of three fourth powers, and no priority claim is made for the displayed families.

The first starting identity is

$$\begin{aligned}(20p^8 - q^8)(2q^7)^4 + (64p^8q - 2q^9)^4 \\ + (64p^9 - 8p^5q^4 - 3pq^8)^4 \\ = (64p^9 + 8p^5q^4 - 3pq^8)^4.\end{aligned} \tag{1}$$

A version of this identity appears in the public Mathematics Stack Exchange discussion cited in [1,2]. It is treated here as a starting example.

The second starting identity is

$$\begin{aligned}-2(190p^8 + 243q^8)(16p^7q^3)^4 + \left[4p^3q(16p^8 + 27q^8)\right]^4 \\ + (32p^{12} - 64p^8q^4 - 81q^{12})^4 \\ = (32p^{12} + 64p^8q^4 + 81q^{12})^4.\end{aligned} \tag{2}$$

This identity has also appeared in the same public discussion [2]. Its exact algebraic derivation is supplied in Section 4.1.

Throughout, p, q are indeterminates. A polynomial identity over $\mathbb{Q}[p, q]$ remains valid after any integer specialization of p, q. Rational choices of template parameters are allowed whenever the stated hypotheses hold; integer coefficients result when the displayed coefficients are integral.

2 A shared factorization

For commuting quantities X, Y, direct expansion gives

$$(X+Y)^4 - (X-Y)^4 = 8XY(X^2+Y^2).$$

The two coefficient-cancellation templates below use the displayed factorization to compare a symmetric pair of fourth powers and then subtract a further fourth power whose coefficients cancel selected terms.

3 First coefficient-cancellation family

Theorem 3.1 (First general template). Let $A, G, R, S, T, K \in \mathbb{Q}$ with $AK \neq 0$, and suppose

$$R^4 = 8GA^3, \qquad T = -\frac{3G^2}{A}, \qquad S = -\frac{2RG^2}{A^2}.$$

Define

$$N_1(p,q) = \frac{40G^7}{A^3K^4}p^8 - \frac{128G^9}{A^5K^4}q^8.$$

Then

$$N_1(p,q)(Kq^7)^4 + [q(Rp^8+Sq^8)]^4 + [p(Ap^8+Tq^8-Gp^4q^4)]^4 = [p(Ap^8+Tq^8+Gp^4q^4)]^4.$$

Proof. Put

$$X = p(Ap^8+Tq^8), \qquad Y = Gp^5q^4.$$

Then the last two fourth powers in the template are $(X-Y)^4$ and $(X+Y)^4$. By the factorization above,

$$(X+Y)^4 - (X-Y)^4 = 8GA^3p^{32}q^4 + (24GA^2T + 8G^3A)p^{24}q^{12} + (24GAT^2 + 8G^3T)p^{16}q^{20} + 8GT^3p^8q^{28}.$$

On the other hand,

$$[q(Rp^8+Sq^8)]^4 = R^4p^{32}q^4 + 4R^3Sp^{24}q^{12} + 6R^2S^2p^{16}q^{20} + 4RS^3p^8q^{28} + S^4q^{36}.$$

The first displayed condition cancels the $p^{32}q^4$ coefficient in the difference. Substituting the second and third displayed conditions cancels the $p^{24}q^{12}$ and $p^{16}q^{20}$ coefficients. The remaining expression is

$$(X+Y)^4 - (X-Y)^4 - [q(Rp^8+Sq^8)]^4 = \frac{40G^7}{A^3}p^8q^{28} - \frac{128G^9}{A^5}q^{36}.$$

The right-hand side equals $N_1(p,q)(Kq^7)^4$ by the definition of N_1, proving the first template.

3.1 Recovering the first starting identity

Take

$$(A, G, R, S, T, K) = (64, 8, 64, -2, -3, 2).$$

These values satisfy the stated conditions, and the definition of N_1 gives $N_1 = 20p^8 - q^8$. The middle base is $q(64p^8 - 2q^8) = 64p^8q - 2q^9$, while the symmetric pair is

$$p(64p^8 - 3q^8 \mp 8p^4q^4) = 64p^9 - 3pq^8 \mp 8p^5q^4.$$

Thus the first template becomes (1).

3.2 A further explicit instance

The choice

$$(A, G, R, S, T, K) = (4, 8, 8, -64, -48, 2)$$

also satisfies the stated conditions. It gives

$$\begin{aligned}(81920p^8 - 1048576q^8)(2q^7)^4 + (8p^8q - 64q^9)^4 \\ + (4p^9 - 8p^5q^4 - 48pq^8)^4 \\ = (4p^9 + 8p^5q^4 - 48pq^8)^4.\end{aligned}$$

This is a valid instance of Theorem 3.1. Its status relative to all prior literature has not been determined, so no claim of novelty is attached to it.

4 Second coefficient-cancellation family

Theorem 4.1 (Second general template). Let $U, V, W, H, J, L \in \mathbb{Q}$ with $L \neq 0$, and suppose

$$H^4 = 8U^3V, \qquad J^4 = 8UW^3, \qquad HJ^3 = 6UVW^2.$$

Define

$$N_2(p, q) = \frac{8U^3W + 8UV^3 - 4H^3J}{L^4}p^8 + \frac{24UV^2W - 6H^2J^2}{L^4}q^8.$$

Then

$$\begin{aligned}N_2(p, q)(Lp^7q^3)^4 + [p^3q(Hp^8 + Jq^8)]^4 + [Up^{12} - q^4(Vp^8 + Wq^8)]^4 \\ = [Up^{12} + q^4(Vp^8 + Wq^8)]^4.\end{aligned}$$

Proof. Set

$$X = Up^{12}, \qquad Y = q^4(Vp^8 + Wq^8), \qquad B = p^3q(Hp^8 + Jq^8).$$

Expanding the factorization gives

$$\begin{aligned}(X + Y)^4 - (X - Y)^4 = 8U^3Vp^{44}q^4 + (8U^3W + 8UV^3)p^{36}q^{12} \\ + 24UV^2Wp^{28}q^{20} + 24UVW^2p^{20}q^{28} \\ + 8UW^3p^{12}q^{36}.\end{aligned}$$

Also,

$$B^4 = H^4p^{44}q^4 + 4H^3Jp^{36}q^{12} + 6H^2J^2p^{28}q^{20} + 4HJ^3p^{20}q^{28} + J^4p^{12}q^{36}.$$

The stated relations cancel the coefficients of $p^{44}q^4$, $p^{20}q^{28}$, and $p^{12}q^{36}$ in the difference. The remaining terms are

$$(X+Y)^4 - (X-Y)^4 - B^4 = (8U^3W + 8UV^3 - 4H^3J)p^{36}q^{12} + (24UV^2W - 6H^2J^2)p^{28}q^{20}.$$

By the definition of N_2, this is $N_2(p,q)(Lp^7q^3)^4$, proving the template.

4.1 Recovering the second starting identity

Choose

$$(U,V,W,H,J,L) = (32,64,81,64,108,16).$$

The three conditions in the theorem hold. Substitution in the definition of N_2 yields

$$N_2(p,q) = -380p^8 - 486q^8 = -2(190p^8 + 243q^8).$$

Furthermore,

$$B = 4p^3q(16p^8 + 27q^8), \qquad Y = q^4(64p^8 + 81q^8).$$

Consequently, the two symmetric bases $X \mp Y$ are

$$32p^{12} \mp (64p^8q^4 + 81q^{12}),$$

and the template is exactly (2). This proves the second starting identity as a polynomial identity, not merely at selected integer values of p, q.

5 A simpler binomial family

The next construction follows directly from the binomial theorem. It is a useful generator of examples, but its elementary nature should be kept in view.

Theorem 5.1 (Binomial-expansion family). For integers λ, μ, r, define

$$P_{\lambda,\mu,r}(p,q) = 4(\mu-\lambda)p^{12} + 6(\mu^2-\lambda^2)p^8q^4 + 4(\mu^3-\lambda^3)p^4q^8 + (\mu^4-\lambda^4-r^4)q^{12}.$$

Then

$$P_{\lambda,\mu,r}(p,q)q^4 + (rq^4)^4 + (p^4+\lambda q^4)^4 = (p^4+\mu q^4)^4.$$

Proof. For an indeterminate t,

$$(p^4+tq^4)^4 = p^{16} + 4tp^{12}q^4 + 6t^2p^8q^8 + 4t^3p^4q^{12} + t^4q^{16}.$$

Subtract the expansion with $t=\lambda$ from that with $t=\mu$, and then subtract $(rq^4)^4 = r^4q^{16}$. The result is exactly $P_{\lambda,\mu,r}(p,q)q^4$, proving the identity.

5.1 Three specializations

For $(\lambda, \mu, r) = (-4, 2, 6)$,

$$P_{-4,2,6} = 24(p^4 - 4q^4)(p^8 + p^4q^4 + 16q^8),$$

so

$$24(p^4 - 4q^4)(p^8 + p^4q^4 + 16q^8)q^4 + (6q^4)^4 + (p^4 - 4q^4)^4 = (p^4 + 2q^4)^4.$$

This specific family appears in the public discussion [2], where it is related to a rescaled Gerardin-type identity.

For $(\lambda, \mu, r) = (1, 3, 2)$,

$$P_{1,3,2} = 8p^{12} + 48p^8q^4 + 104p^4q^8 + 64q^{12},$$

and therefore

$$\begin{aligned}(8p^{12} + 48p^8q^4 + 104p^4q^8 + 64q^{12})q^4 + (2q^4)^4 \\ + (p^4 + q^4)^4 = (p^4 + 3q^4)^4.\end{aligned}$$

For $(\lambda, \mu, r) = (-2, 1, 3)$,

$$P_{-2,1,3} = 12p^{12} - 18p^8q^4 + 36p^4q^8 - 96q^{12},$$

and therefore

$$(12p^{12} - 18p^8q^4 + 36p^4q^8 - 96q^{12})q^4 + (3q^4)^4 + (p^4 - 2q^4)^4 = (p^4 + q^4)^4.$$

6 Interpretation and limitations

1. The first two theorems give parameterized polynomial identities. Their proofs establish exact coefficient cancellation for indeterminates p, q.

2. In (1), (2), and the generated examples, the coefficient $N(p, q)$ varies with the parameters. These identities therefore do not by themselves produce infinitely many solutions for a single fixed integer n in $na^4 + b^4 + c^4 = d^4$.

3. The binomial family is a formal expansion identity. It generates examples, but it is not presented as a difficult or necessarily new theorem.

4. Algebraic validity, primitivity of an integer solution, infinitude for a fixed coefficient, and novelty are distinct questions. The identities proved here address algebraic validity; no complete classification or priority claim is made.

7 Concluding remarks

The factorization $(X+Y)^4 - (X-Y)^4 = 8XY(X^2+Y^2)$ leads to two coefficient-cancellation templates for fourth powers with a variable polynomial coefficient. The two starting identities follow from particular parameter choices, while the binomial family gives an independent elementary source of examples. Further work could classify parameter choices up to scaling and substitution, determine when the resulting families are inequivalent, and investigate separately whether a specialization yields infinitely many solutions for a fixed coefficient n.

References

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